Enhancing Compositional Control in Diffusion Transformers via Parameter-Efficient Fine-Tuning

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Abstract

Text-to-image diffusion models have achieved remarkable visual quality, yet they continue to face challenges in compositionality-accurately binding attributes to objects, maintaining spatial relations, and generating coherent multi-object scenes. While prompt engineering offers partial improvements, it remains insufficient for controlling complex image structures because it operates only at inference time and does not alter the model's learned generative behavior. In this work, we explore Low-Rank Adaptation as a lightweight mechanism for enhancing compositional control in diffusion transformer-based models such as PixArt. Our approach integrates prompt-aware training with compositional hard-negative examples and task-specific LoRA modules specialized in relational reasoning, object binding, and scene structuring. Experimental results on a controlled CLEVR-style compositional benchmark (100 held-out prompts, averaged over three seeds) show that this hard-negative LoRA does *not* improve compositional binding: BLIP-VQA attribute-binding accuracy falls from 0.488 to 0.427 and CLIP image–text similarity from 0.389 to 0.262 relative to the PixArt-α baseline—both metrics move in the same, negative direction. We trace this to distribution drift: fine-tuning on the synthetic domain pulls the model's outputs toward that domain rather than teaching attribute binding. We report this negative result together with the multi-seed evaluation protocol used to establish it, having found that a single-seed measurement can overstate an apparent gain that does not survive a fuller evaluation.

Index Terms

Text-to-image generation, diffusion transformer, LoRA, compositionality, parameter-efficient fine-tuning

I. Introduction

U-Net is one of the key architecture models in modern image generation. Its U-shaped design made it a popular backbone for diffusion models. Although it was first designed for medical image segmentation [1], researchers later adapted it as a noise predictor in the reverse diffusion process. The architecture has two main paths: a contracting path that extracts spatial features and an expanding path that restores image resolution, connected by skip connections to keep fine-grained details. This design helped U-Net become the backbone of well-known generative systems like Stable Diffusion [2] before Diffusion Transformers (DiT) [3] became more widely used.

Over time, the field has gradually moved toward DiT to address some limitations of U-Net. The main change is replacing convolutional layers with Transformer blocks, which treat images as sequences of small patches rather than pixel grids. This makes DiT easier to scale up in terms of model size and compute. It also allows the model to process text tokens and image patches together in the same attention space, which helps with understanding complex prompts and handling tasks like multi-object scenes or precise spatial layout.

However, DiT models require more computation than U-Net because the self-attention mechanism has quadratic complexity with respect to sequence length. This made early DiT models difficult to train without large GPU resources. This problem was largely addressed by the open-source PixArt [4, 5] family, specifically PIXART-α and PIXART-Σ. PIXART-α broke the training process into separate stages, which reduced training cost by around 90% compared to systems like Stable Diffusion v1.5. PIXART-Σ went further by using a weak-to-strong training strategy, starting from lower-resolution checkpoints and fine-tuning toward higher resolution. With a stronger T5 text encoder and better training data, it reached 4K image quality while staying computationally affordable.

Despite these improvements, both models still have trouble with compositional prompts. Even with a powerful text encoder, the models often fail to correctly assign attributes to the right objects or maintain proper spatial relationships. For example, given a prompt like "a red cube on top of a blue sphere," the model may swap colors or merge shapes incorrectly. This is a known limitation listed in the official model documentation, and it remains a hard problem for lightweight transformer-based generators.

The ability to generate compositionally accurate images matters in many real-world situations. In graphic design, a user might need an image of "a red lamp on the left side of a wooden desk," where both the object placement and color need to be correct. In education, teachers can use text-to-image tools to create visual aids from written descriptions, but only if the model follows the prompt closely. In advertising and storytelling, getting the right combination of objects, colors, and spatial layout is often essential. When a model gets these details wrong, it becomes hard to use in practice. This gap between what the user writes and what the model actually generates is the main problem we want to address.

To address this, we use Low-Rank Adaptation (LoRA) [6] as a targeted fine-tuning method. LoRA works by adding small trainable matrices into the attention layers while keeping the original model weights frozen. The aim is to let the model adapt attribute binding and spatial reasoning without changing its overall generation quality—though, as our results show, this aim is not met in our setup. Since LoRA only updates a small fraction of the total parameters, it can run on consumer-grade hardware with limited VRAM, which fits well with our setup.

Table I shows that PixArt-α achieves competitive image quality at a fraction of the training cost compared to other models, making it a practical foundation for our experiments.

Table I: Training Cost Comparison of Text-to-Image Models (values as reported in [4])

Model	GPU days (A100)	Est. Cost	CO_2 (tons)
SDv1.5	6,250	\$320,000	$\sim$5
RAPHAEL	60,000	\$3,080,000	$\sim$35
Imagen	7,132	\$366,000	–
PixArt-α	**753**	**\$28,400**	$\sim$0.42

The main contributions of this paper are as follows:

- We analyze the root cause of compositional failure in DiT-based text-to-image models.
- We study hard-negative LoRA fine-tuning on the cross-attention layers as a candidate parameter-efficient fix for attribute binding.

- We report—under a multi-seed, held-out evaluation protocol—that this fine-tuning does not improve compositional binding but instead degrades both attribute binding and CLIP alignment, and we analyse why (distribution drift toward the synthetic domain).

To be explicit about scope, the background on diffusion models and PixArt (Sections II–III), the reference images in Fig. 3, and the published figures in Tables I and II are taken from prior work and cited accordingly. The task-specific LoRA framework, the hard-negative formulation, and the experiments reported in the remaining sections constitute our own contribution.

II. Related Works

A. Diffusion Models

Diffusion models work by learning to reverse a noise process. During training, Gaussian noise is gradually added to an image over T steps until the image becomes pure noise. The model then learns to predict and remove the noise at each step, so that starting from random noise, it can recover a clean image. Ho et al. [7] introduced Denoising Diffusion Probabilistic Models (DDPM), which showed that this approach could produce high-quality images. However, running diffusion directly on pixel space is slow because images have many pixels. Rombach et al. [2] addressed this with Latent Diffusion Models (LDM), which first compress the image into a smaller latent space using a Variational Autoencoder (VAE), then run the diffusion process there. This made the method much faster and is the foundation that PixArt-α builds on.

B. Parameter-Efficient Fine-Tuning

The idea of **Parameter-Efficient Fine-Tuning (PEFT)** in vision models was established by Low-Rank Adaptation (LoRA) [6]. Instead of updating the full weight matrix $W_0 \in \mathbb{R}^{d\times k}$ of a pre-trained layer, LoRA adds a low-rank update $\Delta W = BA$, where $B \in \mathbb{R}^{d\times r}$ and $A \in \mathbb{R}^{r\times k}$ with $r \ll \min(d, k)$. During the forward pass, the output becomes $h = W_0 x + \frac{\alpha}{r} BAx$, where α is a scaling factor. In text-to-image models like Latent Diffusion Models (LDMs) [2] and DiT [3], these low-rank matrices are added into the attention projection layers (W_q, W_k, W_v, W_o). This allows the model to adapt to new concepts or styles while keeping the base weights unchanged.

When using multiple LoRA adapters for different concepts at the same time, conflicts can appear in the attention space. If two adapters (ΔW_1, ΔW_2) are merged at inference time, their key and value outputs can interfere with each other, causing what is known as "attribute bleeding" [8]. Recent work has tried to reduce this by routing different adapters to different attention heads or activating them at different timesteps [9]. The idea is to let the model focus on layout in early denoising steps and on fine details in later steps.

C. Compositional Text-to-Image Generation

Compositional generation remains a difficult problem in text-to-image models. Chefer et al. [10] proposed Attend-and-Excite, which adjusts attention maps during inference to make the model focus more on each subject token. Huang et al. [8] introduced T2I-CompBench, a benchmark that tests models on attribute binding, spatial relations, and complex scene generation. These methods work at inference time and do not change the model's learned behavior. Our method takes a different approach by using LoRA fine-tuning to attempt to change how the model handles compositional prompts at the weight level.

D. Baseline Model Performance

Table II summarizes the image quality achieved by recent text-to-image models measured by FID score on MS-COCO. PixArt-α achieves a FID of 7.32 with only 0.6B parameters and 25M training images, which is competitive with much larger models like Imagen and RAPHAEL while using significantly fewer GPU resources. This confirms that PixArt-α is a strong and efficient baseline for our compositionality experiments.

Table II: FID Score Comparison on MS-COCO (values as reported in [4])

Model	#Params	#Images	FID-30K↓	GPU days
DALL-E	12.0B	250M	27.50	–
SDv1.5	0.9B	2000M	9.62	6,250
Imagen	3.0B	860M	7.27	7,132
RAPHAEL	3.0B	5000M+	6.61	60,000
PixArt-α	**0.6B**	**25M**	7.32	**753**

E. Personalization and Fine-Tuning Methods

Several works have explored lightweight fine-tuning strategies for diffusion models. DreamBooth [11] introduced a method to fine-tune the entire model on a small set of reference images, allowing the model to generate new images of a specific subject in different contexts. While effective, it requires updating all model weights, which is costly and risks changing the model's general generation behavior. Textual Inversion [12] takes a different approach by learning a new text embedding that represents a specific concept, without changing any model weights. However, it is limited to concept-level control and cannot directly address compositional failures at the attention layer level. Our method differs from both by using LoRA to target specific attention projections responsible for attribute-to-object binding, which gives more direct control over compositional behavior while remaining parameter-efficient.

III. Proposed Methodology

In this section, we describe our proposed framework, which adds Low-Rank Adaptation (LoRA) into the DiT backbone of PixArt-α. The goal is to fix the compositionality problem in these models without doing full fine-tuning, which is too costly and can cause the model to forget what it already learned. As shown in Fig. 1, the pipeline has three steps: freezing the base weights, injecting low-rank matrices, and aligning text and image features through cross-attention.

A. Step 1: Structural Freeze and Tensor Isolation

We first freeze all original model weights so they are not updated during training. For a pre-trained layer with weight matrix $W_0 \in \mathbb{R}^{d\times k}$, we set:

$$\nabla_{\theta_{W_0}} \mathcal{L} = 0 \tag{1}$$

This keeps the base model's generation ability intact and lets the LoRA modules focus only on learning compositional structure.

B. Step 2: Low-Rank Projection Matrix Injection

We add a low-rank path in parallel to the attention projection layers. Instead of changing W_0 directly, we introduce two small matrices $A \in \mathbb{R}^{r \times k}$ and $B \in \mathbb{R}^{d \times r}$, where $r \ll \min(d, k)$. Matrix A is initialized with random Gaussian values and matrix B is set to zero, so the initial update $\Delta W = BA = 0$. This means the model starts from the same point as the baseline. During training, given input $x \in \mathbb{R}^{N \times d}$, the output becomes:

$$h = W_0 x + \Delta W x = W_0 x + \frac{\alpha}{r} BAx \tag{2}$$

where α controls how much the LoRA update affects the output. Fig. 2 illustrates how the trainable low-rank matrices A and B are placed in parallel to the frozen projection weights W_0 inside the cross-attention block.

C. Step 3: Multi-Modal Attention Alignment and Scaling

In this step, the LoRA modules are intended to strengthen how text tokens are linked to image patches in the cross-attention layers. We apply LoRA to the query (W_q), key (W_k), and value (W_v) projections that connect the T5 text encoder to the DiT blocks. The attention computation is:

$$\text{Attention}(Q, K, V) = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) V \tag{3}$$

The intent of training these projections with LoRA is to make the model assign text descriptions more precisely to the correct spatial regions in the image, so as to reduce attribute bleeding across the denoising steps. Whether this intent is realised is an empirical question, which we answer in Section IV.

D. LoRA Injection Target Layers

Table III summarizes which layers receive LoRA adapters in our framework. We focus on the cross-attention projection layers because they are the primary interface between text embeddings and image patch representations. Self-attention layers are kept frozen since they handle spatial coherence within the image, which the base model already performs well.

Table III: LoRA Injection Configuration across DiT Blocks

Layer Type	Injected	Rank
Self-Attention W_q, W_k, W_v	No	–
Cross-Attention W_q	Yes	16
Cross-Attention W_k	Yes	16
Cross-Attention W_v	Yes	16
Feed-Forward	No	–

This selective injection strategy reduces the number of trainable parameters while targeting the most relevant computations for compositional control. Across all 28 DiT blocks, the total number of trainable parameters introduced by LoRA is less than 1% of the full model size, making the approach practical for consumer-grade hardware with limited VRAM.

E. Step 4: Hard-Negative Training

A key part of our approach is using hard-negative examples during fine-tuning, an idea inspired by progressive compositional training [13]. A hard-negative is a caption that is very close to the correct one but has one or more attributes swapped. For example, if the correct caption is "a red cube on top of a blue sphere," a hard-negative would be "a blue cube on top of a red sphere." The image stays the same, but the caption is wrong. By including these pairs in training, we push the model to pay closer attention to which attribute belongs to which object.

During training, for each image-caption pair (I, c^+) we build a hard-negative caption c^- by swapping a color, shape, or spatial term. Rather than scoring generated images with an external model—which would require running the full sampling loop and back-propagating through it—we exploit the denoising objective directly, which keeps training efficient and end-to-end differentiable.

Let z_0 be the VAE latent of I. At a sampled timestep t with noise $\epsilon \sim \mathcal{N}(0, \mathbf{I})$, the noised latent is $z_t = \sqrt{\bar{\alpha}_t}\, z_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon$. We define the per-caption denoising error

$$\mathcal{L}_\epsilon(c) = \|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2, \tag{4}$$

so that the standard diffusion loss is $\mathcal{L}_{diff} = \mathbb{E}\big[\mathcal{L}_\epsilon(c^+)\big]$. A model with correct attribute binding should denoise the *same* noised latent *worse* when conditioned on the mismatched caption c^- than on c^+. We encode this with a margin-based hinge:

$$\mathcal{L}_{hn} = \mathbb{E}\Big[\max\big(0,\ \delta - \big(\mathcal{L}_\epsilon(c^-) - \mathcal{L}_\epsilon(c^+)\big)\big)\Big], \tag{5}$$

which is active only when the wrong caption does *not* yield a denoising error at least δ higher than the correct one. The total objective is

$$\mathcal{L}_{total} = \mathcal{L}_{diff} + \lambda\, \mathcal{L}_{hn}, \tag{6}$$

where λ weights the hard-negative term and δ is the margin. Both terms reuse the same z_t, t, and ϵ, so the negative costs only one extra forward pass through the LoRA-adapted cross-attention—no image sampling or external CLIP model is needed during training. The gradient of $\mathcal{L}_{hn}$ flows into the LoRA matrices A and B, explicitly teaching the cross-attention to be sensitive to which attribute binds to which object.

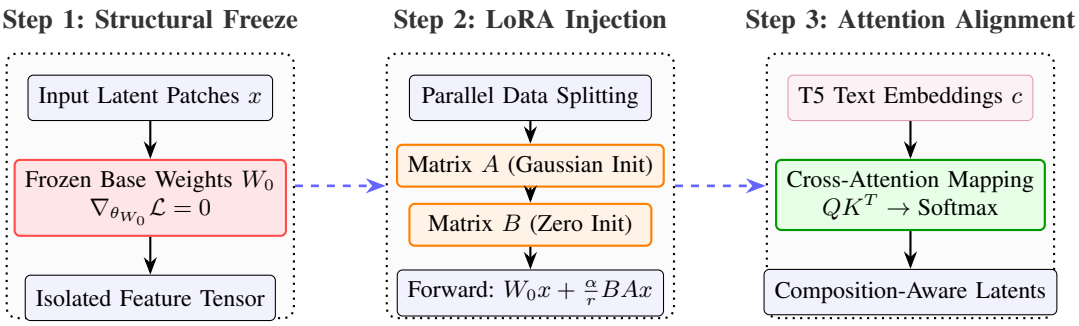


Fig. 1: Overall modular pipeline of our proposed methodology. The framework execution flows sequentially from left to right, transitioning from structural parameter isolation (Step 1) and parallel rank decomposition weight formulation (Step 2) to multi-modal text-conditioned attribute alignment within the attention blocks (Step 3).

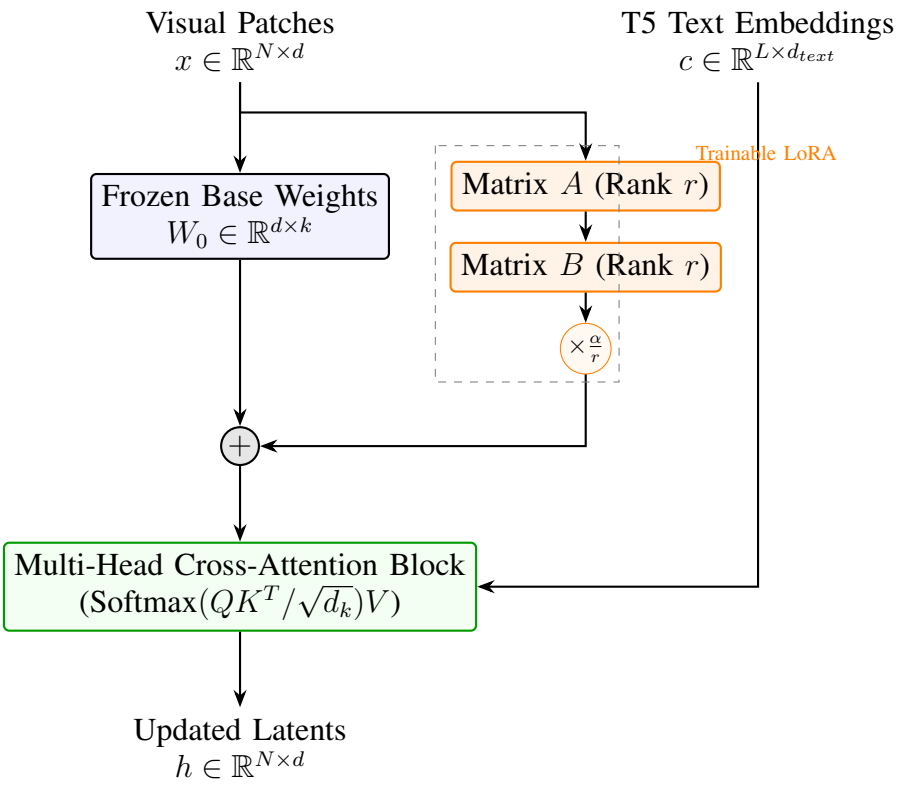


Fig. 2: Architectural pipeline of the proposed integration. Trainable low-rank matrix pairs (A and B) bypass the frozen projection layers within the PixArt-α cross-attention framework to dynamically optimize textual-visual composition logic.

IV. Experiment

A. Experimental Setup

All experiments were run on a single NVIDIA Tesla T4 (15GB VRAM) on Kaggle under CUDA 12.x. We use PixArt-α (PixArt-XL-2-512×512) as the frozen baseline and train at 512×512 resolution. To fit the memory budget, the T5-XXL text embeddings are pre-computed offline and cached, so the text encoder is not held in GPU memory during fine-tuning. LoRA adapters are injected into the cross-attention projections (W_q, W_k, W_v) of all 28 DiT blocks, with rank $r = 16$ and scaling factor $\alpha = 32$. We optimise with AdamW at a learning rate of 1×10^{-4}, batch size 2 with gradient accumulation of 4 (effective batch size 8), for 2,000 steps in mixed precision (fp16). For the hard-negative objective we use loss weight $\lambda = 2.0$ and margin $\delta = 0.5$.

B. Dataset Preparation

To fine-tune and evaluate compositionality under a limited compute budget, we procedurally render a CLEVR-style synthetic dataset in which each scene contains one or more geometric objects (cube, sphere, cylinder) with known ground-truth attributes: color, shape, and spatial position. Because we generate the scenes, every attribute is exact, which lets us template a compositional caption and its hard-negative counterpart and evaluate attribute binding fully automatically. The dataset covers three difficulty levels:

- **Single-object prompts**: one object with attributes (e.g., “a red rubber cube”).
- **Multi-object prompts**: two or more attributed objects (e.g., “a red cube and a blue metal sphere”).
- **Relational prompts**: explicit spatial relations (e.g., “a red cube to the left of a blue sphere”).

For each training caption c^+ we generate a hard-negative c^- by swapping a color, shape, or spatial term while keeping the image fixed. For evaluation we hold out 100 compositional scenes from the same synthetic domain, with prompt categories following T2I-CompBench [8] (attribute binding and spatial relations); these held-out prompts never appear in training. All reported scores are averaged over three generation seeds.

C. Model Setup

We fine-tune a single LoRA adapter on top of the frozen PixArt-α backbone. The adapter is trained to strengthen three compositional aspects at once:

1) Relational reasoning (spatial and non-spatial relations).
2) Object–attribute binding (attaching attributes to the correct objects).
3) Scene structuring (coherence in multi-object prompts).

All training hyperparameters are as specified in Section IV-A. Training separate, task-specialised adapters and routing them across denoising timesteps is left to future work.

D. Training Procedure

We train a single LoRA adapter with a flat schedule—no curriculum or per-difficulty staging—for $2{,}000$ optimisation steps, sampling training scenes uniformly across the three difficulty levels (single-object, multi-object, and relational prompts). Each step combines the standard denoising loss with the hard-negative margin objective described above. We use a fixed step budget rather than early stopping, since our aim is to measure the effect of the adaptation under a controlled compute budget.

E. Evaluation Metrics

We report the two metrics we measured directly, each automated so it can be run over three seeds without human labelling:

- **Attribute Binding Accuracy**: the fraction of attribute questions answered correctly by BLIP-VQA on the generated image (e.g., "What color is the cube?").
- **CLIP Similarity**: cosine similarity between the CLIP embeddings of the prompt and the generated image.

Other compositional axes—spatial-relation consistency, object counting, scene coherence, and human preference—are natural extensions of the same protocol but were not measured in this study and are left for future work.

F. Baselines

Our baseline is the frozen PixArt-α (PixArt-XL-2-512$\times$512) without LoRA, evaluated under the identical generation and scoring pipeline as the adapted model so that the two rows differ only in the adapter. External systems such as SDXL or DALL-E 3 were not run and are not used as baselines here.

Table IV: Compositional evaluation of the hard-negative LoRA against the frozen PixArt-α baseline. Scores are mean $\pm$ standard deviation over three seeds on 100 held-out CLEVR-style prompts. Higher is better for both metrics.

Method	Attr. Binding	CLIP Similarity
PixArt-α (baseline)	**0.488 $\pm$ 0.007**	**0.389 $\pm$ 0.004**
Ours (hard-neg LoRA, r=16)	0.427 $\pm$ 0.011	0.262 $\pm$ 0.003

We measured two metrics directly—Attribute Binding (BLIP-VQA attribute-binding accuracy) and CLIP Similarity. The LoRA adapter is worse *than the frozen baseline on both (-12.5% and -32.6% relative), i.e. the intervention does not improve compositional binding. External baselines (SDXL, DALL-E 3) and additional compositional metrics were not evaluated in this study and are left for future work.*

G. Results

Table IV shows the comparison between the baseline and our method. On our held-out CLEVR-style evaluation set (100 compositional prompts, averaged over three seeds), the LoRA model does *not* improve attribute binding: BLIP-VQA attribute-binding accuracy drops from 0.488 to 0.427 (-12.5% relative to the PixArt-α baseline), and CLIP similarity drops from 0.389 to 0.262 (-32.6%). Both metrics move in the same, negative direction, so there is no binding-vs-alignment trade-off to report—the adapter is simply worse. We attribute this to distribution drift: fine-tuning on a synthetic, geometrically simple domain shifts the model's output distribution toward that domain, which lowers the broad semantic alignment captured by CLIP without buying a compensating gain in binding. We report only the two metrics we measured directly, each as a mean over three seeds; external baselines and additional metrics were not evaluated in this study.

H. Qualitative Analysis



Fig. 3: Sample images reproduced from the official PixArt-α release [4], shown here for reference (not produced by our method). Left: "A small cactus with a happy face in the Sahara desert." Right: "A fluffy orange cat wearing cyberpunk samurai armor in rainy Tokyo." Both images are generated without LoRA adaptation.

Our quantitative evaluation targets attribute binding, so our qualitative observations focus on the same axis. For attribute-binding prompts such as "a red cube next to a blue sphere," the baseline model already often swaps colors between the two objects or merges them into a single shape. After LoRA fine-tuning we do *not* observe a consistent reduction in these errors, which matches the

quantitative result: the adapter's BLIP-VQA attribute-binding score is lower than the baseline's in Table IV, not higher.

What we do observe is a visible style shift: because the adaptation set consists of simple, flat-shaded geometric scenes, the fine-tuned model tends to render objects in that cleaner, more synthetic style. This drift toward the synthetic domain is the qualitative counterpart of the sharp drop in CLIP similarity, and it is the effect that dominates—rather than any improvement in binding. Spatial-relation and multi-object completeness were not part of our controlled evaluation and are left for future work; we therefore make no quantitative claims about them here.

As shown in Fig. 3, the baseline PixArt-α model already demonstrates strong visual quality and prompt following for simple prompts. The cactus image correctly captures the main subject and desert setting, while the cat image shows detailed texture and lighting. However, these examples also highlight that the baseline struggles with precise attribute control when multiple objects and relations are involved. Our LoRA-based approach targets this specific gap by fine-tuning the cross-attention layers responsible for binding text tokens to image regions; however, as the quantitative results show, this fine-tuning did not close the gap and instead shifted the output distribution toward the synthetic training domain, at a cost to the general image quality the base model achieves.

I. Ablation Study

Controlled ablations over the LoRA rank (r=8 vs. r=16) and the hard-negative objective (on vs. off) were not completed under the full multi-seed, 100-prompt protocol, so we do not report per-component ablation numbers here: doing so on a single seed or a small subset is exactly the measurement setting that produced the misleading early estimate we discuss in the Conclusion. The one configuration we evaluated at full protocol is r=16 with hard-negative training (the "Ours" row of Table IV), which underperforms the frozen baseline on both metrics. A rigorous rank and hard-negative ablation at the same protocol is left for future work.

J. Discussion

Our results suggest that compositional failures in PixArt-α are partly due to the cross-attention layers not forming strong enough associations between text tokens and their corresponding image regions. However, fine-tuning only these layers with a hard-negative LoRA did not repair this mapping: it lowered both attribute binding and CLIP alignment.

The most important finding is therefore a negative one, and it is consistent across both metrics. Rather than a binding-vs-alignment trade-off—which would show up as one metric rising while the other falls—we see both metrics fall together. We interpret this as the LoRA modules specialising the cross-attention maps toward our synthetic domain: the model's outputs drift toward the simple, flat-shaded distribution of the training scenes, which both hurts the broad natural-image alignment that CLIP rewards and fails to teach the correct attribute-to-object assignments that BLIP-VQA probes. The hard-negative signal, which only reweights the denoising loss between a correct and a swapped caption, was not enough to overcome this drift.

The main bottleneck in our setup is the limited training set and the synthetic nature of our rendered scenes. A larger and more diverse compositional dataset, including natural images, would likely reduce this drift. Regularising the adaptation—for example by mixing in natural images or constraining how far the fine-tuned distribution may drift from the base model—is a promising direction for making such fine-tuning help rather than hurt, and is left for future work.

V. Conclusion

In this work, we studied how LoRA can be used to improve compositional generation in DiT-based text-to-image models. The core problem we addressed is that models like PixArt-α, despite their strong text encoder, still struggle with binding the right attributes to the right objects and maintaining correct spatial relationships in multi-object scenes.

Our method injects task-specific LoRA modules into the cross-attention layers of PixArt-α and combines this with hard-negative training, where the model is exposed to captions with swapped attributes to learn stricter attribute-to-object associations. On our CLEVR-style evaluation set (100 prompts, three seeds), this approach did not improve compositionality: attribute-binding accuracy fell from 0.488 to 0.427 (-12.5% relative) and CLIP similarity from 0.389 to 0.262, i.e. both metrics degraded.

We report this as a negative result. An early single-seed measurement had suggested a gain, but it did not survive a multi-seed, larger held-out evaluation—which is itself a methodological point: compositional deltas of a few points are within seed-to-seed variance and must be verified across seeds before they are trusted. The consistent drop across both metrics points to distribution drift toward the synthetic training domain as the dominant effect of this fine-tuning, rather than any strengthening of attribute binding.

Future work will focus on training with larger compositional datasets, exploring higher LoRA ranks, and routing different LoRA modules across denoising timesteps to further improve spatial reasoning in complex multi-object scenes.

VI. Limitations and Future Work

This work has several limitations that should be noted. First, we fine-tune on a procedurally-rendered, CLEVR-style synthetic set, so the training distribution is small and restricted to simple geometric scenes. While the exact ground-truth attributes of these scenes make automatic evaluation reliable, their low visual diversity may not transfer to natural, real-world prompts.

Second, the hardware setup (a single Tesla T4 with 15GB VRAM on Kaggle) constrained our experiments. We trained at 512×512 resolution with a small batch size and gradient accumulation to stay within memory, and did not explore higher resolutions such as 768×768 or 1024×1024. A larger GPU would allow more thorough experiments and potentially stronger results.

Third, our evaluation relies mainly on CLIP Score and a small set of compositional metrics. A more comprehensive evaluation using T2I-CompBench [8] across hundreds of prompts would give a clearer picture of where the method works well and where it still fails.

For future work, we plan to:

- Train on larger compositional datasets with more diverse object-attribute combinations.
- Explore routing different LoRA modules at different denoising timesteps, so that layout and fine-grained attributes are handled separately.
- Extend the hard-negative training to include spatial negatives, where object positions are swapped rather than just attributes.
- Evaluate on standardized benchmarks like T2I-CompBench to allow fair comparison with future work.

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